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Exp-13: Implement the Car Price Prediction Model using Python

AIM

To implement a Car Price Prediction Model using Python and predict the selling price of a car based on its features such as year, mileage, engine size, and other attributes.

ALGORITHM

1. Import the required Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.
2. Load the car dataset.
3. Display and examine the dataset.
4. Check and handle missing values.
5. Select the relevant features such as Year, Present_Price, Kms_Driven, Fuel_Type, and Selling_Type.
6. Convert categorical values into numerical values.
7. Separate the input features X and target variable Y (car selling price).
8. Split the dataset into training and testing data.
9. Train a Linear Regression model using the training data.
10. Predict car prices using the test data.
11. Calculate the model performance using Mean Squared Error (MSE) and R² score.
12. Give the predicted price for a new car.

PROGRAM

```
# Import required libraries

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset (used as a sample regression dataset)

data = fetch_california_housing()

X = data.data
y = data.target

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the Linear Regression model

model = LinearRegression()

# Train the model
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model.fit(X_train, y_train)

# Predict the test data

y_pred = model.predict(X_test)

# Display results

print("Actual Prices:")

print(y_test[:10])

print("\nPredicted Prices:")

print(y_pred[:10])

# Model Evaluation

print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))

```

Output:

```

... Actual Prices:
[0.477  0.458  5.00001 2.186  2.78  1.587  1.982  1.575  3.4
 4.466  ]

Predicted Prices:
[0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367
 2.64550005 2.16875532 2.74074644 3.91561473]

Mean Squared Error: 0.5558915986952422
R2 Score: 0.5757877060324524

```

SAMPLE INPUT

Feature	Value
Year	2018
Present Price	₹8.50 lakh
Kms Driven	30,000 km
Fuel Type	Petrol
Selling Type	Dealer
Transmission	Manual

SAMPLE OUTPUT

Car Price Prediction

Predicted Selling Price: ₹6.20 lakh

Model Performance:

Mean Squared Error: 1.25

R² Score: 0.89

RESULT

The **Car Price Prediction Model was successfully implemented using Python**. The trained machine-learning model predicts the approximate selling price of a car based on its given features.

